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Author(s): Blaine F. Severin, Makram T. Suidan, Bruce E. Rittmann and Richard S. Engelbrecht

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Inactivation kinetics in a flow-through UV reactor

Blaine F. Severin, Makram T. Suidan, Bruce E. Rittmann, Richard S. Engelbrecht

The disinfection of treated water and wastewater effluents by ultraviolet light (UV) is a viable alternative to chlorination.¹⁻⁹ One interesting aspect of UV inactivation is the relation of rapid kinetics to the effects of various mixing regimes within non-uniform light intensity fields. Most available literature has been confined to the description of results obtained from UV reactors of proprietary design with little effort to describe disinfection results in continuous-flow reactor systems with known mixing character.

Cortelyou *et al.*¹⁰ reported an increase in disinfection efficiency by increasing turbulence in a single-lamp, continuous-flow reactor. The character of the flow regime in the reactor, however was not established. In batch and plug-flow reactors, the mathematics simulating the inactivation of *Escherichia coli* were described as a first order rate with respect to the surviving fraction of organisms³. Oliver and Cosgrove⁵ showed that the rate of inactivation of coliform organisms is first order with respect to UV intensity. Haas and Sakellaropoulos¹¹ used the assumptions of mixed second-order kinetics with respect to both organism density and UV light intensity and presented mathematical solutions describing the efficiency of single lamp reactors subjected to various flow regimes. In addition, there is still an accepted notion that inactivation data can be treated using simple mixed second-order kinetics by ignoring initial resistance in batch data and extracting a kinetic constant from the linear portion of the inactivation curve.

This study examines the inactivation of *Escherichia coli*, bacterial virus f2, and *Candida parapsilosis* in a batch UV reactor and in a completely mixed flow-through reactor. Results indicate that mixed second-order kinetics are not generally applicable as a means of scaling between reactor systems with different mixing regimes.

EQUIPMENT AND PROCEDURES

Continuous flow disinfection data were developed using a completely mixed, single lamp, annular reactor (Figure 1). The reactor was made from a 14.0-cm diameter (ID) plexiglass column, 27 cm high. The volume of the reactor, excluding the volume of the quartz tube, was 4010 cm³. A 2.54-cm diameter (OD) quartz tube was placed through the center of the reactor to form an annulus. The quartz tube housed a low pressure mercury vapor lamp. The lamp was powered by a fluorescent lamp D.C. ballast. Voltage across the ballast was constant at 140 V as supplied by a voltage regulator. The inside wall of the reactor was painted lamp black to minimize reflection at the wall surface. The incident light intensity at the surface of the quartz tube was 34 100 $\mu\text{W}/\text{cm}^2$, measured using the potassium

ferrioxalate chemical actinometry test¹². Complete mixing in the reactor was achieved by the use of two impeller systems. One impeller shaft was placed on the influent side of the lamp and the other on the discharge side of the lamp.

Each impeller shaft contained five, three-blade impellers equally spaced along the shaft. The shafts were driven by motors rated at 6000 rpm and operated at about 85% capacity. Water entered the side of the reactor near the bottom and exited near the top of the reactor on the side opposite the entrance. The reactor was operated under a slight head to eliminate free face vortexing. Flows could be adjusted to provide retention times between 20 and 260 seconds.

Mixed second-order kinetics can give predictive errors between batch and continuous-flow reactors.

Water used in the experiments with the completely mixed flow-through reactor was prepared in a 200-L reservoir by adding 1 L of stock phosphate buffer solution (24 g KH₂PO₄ and 6 g NaOH per litre of stock solution) and appropriate amounts of organism culture and UV light-attenuating agent to deionized (DI) water. The buffer maintained the pH between 6.5 and 7.0. Organism densities in the test water varied between 2×10^5 and 1×10^6 colony forming units (cfu) or plaque forming units (pfu) per mL test water. The UV-attenuating agent was parahydroxybenzoic acid (PHBA), which has a narrow absorbance peak around 244 nm. The absorbance spectra for 5 mg/L and 15 mg/L solutions of PHBA in DI water at wavelengths near 254 nm are presented in Table 1. PHBA was used to simulate natural tannic acid components present in wastewater. Quantities up to 500 mL of a stock solution PHBA (10 g/L) added to 200 L DI water provided a range of absorbances (A) (base 10) at 254 nm between 0.04 and 2.2 cm⁻¹. PHBA in this range of concentrations was not toxic to the microorganisms. The temperature of the test water was $20 \pm 1.0^\circ\text{C}$.

Batch disinfection data were generated using a flat-tray reactor with a 2.54-cm water depth and a total volume of 500 mL. The UV source was a lamp mounted in a parabolic, stainless steel reflector system using ballast and operated at line voltage. The lamp was placed 63 cm above the surface of the water in the tray. A shutter arrangement was used to control exposure time of the test water to UV. A magnetic stirrer provided mixing in the reactor. The UV intensity was measured to be 450 $\mu\text{W}/\text{cm}^2$ at the water surface.

Test water used in the batch disinfection experiments was prepared by adding 2.5 mL of stock phosphate buffer solution

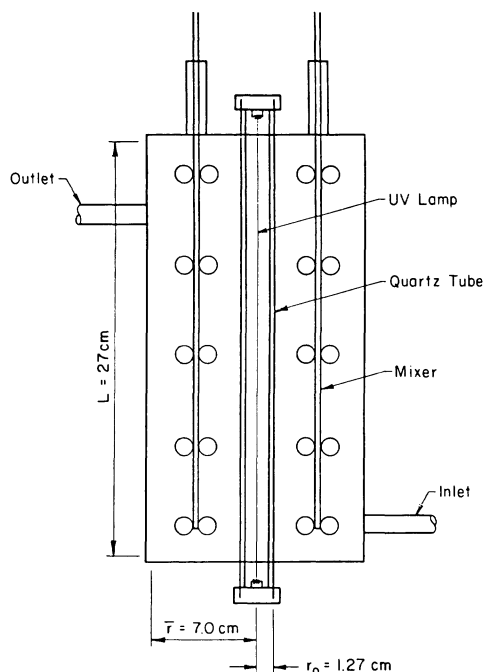


Figure 1—Completely mixed, flow-through UV reactor.

and 500 mL DI water to the batch reactor. This solution was sterilized for 5 minutes under the UV lamp prior to the addition of the appropriate organism culture. The organism densities ranged from 2×10^5 to 1×10^6 cfu or pfu per mL test water. The pH of the water was between 6.5 and 7.0, and the temperature of the water was $20 \pm 1.0^\circ\text{C}$. No PHBA was added to the water and in all tests the UV absorbance was below $0.04\text{ (cm}^{-1}\text{)}$. At this level, the average intensity through the water column was approximately 95% of the input energy flux at the water surface.

A pure culture of *C. parapsilosis* was isolated by Engelbrecht *et al.*¹³ and maintained on refrigerated slants of yeast malt agar (YMA). YMA is prepared from 3 g yeast extract, 3 g malt, 5 g peptone, 10 g dextrose, and 12 g agar in 1 L DI water. Disinfection tests were performed on 20 to 24 hour cultures of *C. parapsilosis* grown in yeast malt broth (YMA without agar) on a shaking waterbath at 27°C . The yeast was enumerated using pour plates of YMA incubated for 3 days at 27°C .

A pure culture of *E. coli*, initially isolated from a secondary wastewater effluent, was maintained on refrigerated nutrient agar slants. Fresh cultures of *E. coli* were grown at 37°C on a shaker bath in nutrient broth with 100 ppm by volume of polyoxyethylene sorbitan mono oxylate, a nonionic surfactant, added as an anticlumping agent. Cultures grown for 20 to 24 hours

Table 1—Absorbance spectrum for parahydroxybenzoic acid (PHBA) in DI water.

Wavelength λ (nm)	Absorbance A (cm ⁻¹)						
	229	234	244	254	264	274	279
5 mg/L PHBA	0.214	0.293	0.426	0.352	0.161	0.070	0.036
15 mg/L PHBA	0.620	0.860	1.252	1.037	0.450	0.198	0.102

were used in the disinfection tests. *E. coli* was enumerated using 20 to 24 hour nutrient agar pour plates incubated at 35°C .

Initial cultures of f2 virus and host organism, *E. coli* K-13, were obtained from Vincent Oliveri at the Johns Hopkins University, Baltimore, Md. Bacterial virus f2 was cultured and enumerated using a modified method described by Leob and Zinder¹⁴. Host cultures of *E. coli* K-13 were grown on TYE broth (tryptone 10 g/L, yeast extract 1 g/L, and sodium chloride 8 g/L) at 37°C on a shaking water bath. Stock suspensions of f2 virus were prepared by the addition of phage (0.1 infectivity ratio of virus to bacteria) to pregrown cultures of *E. coli* K-13 at a density of approximately 3×10^8 bacteria/mL. After the infected culture lysed (8 to 12 hour incubation at 37°C on a shaking water bath), cell debris was removed by centrifugation at 2500 rpm for 15 minutes. The supernatant liquor was used as the virus stock and was maintained in 5 mL aliquots at 4°C . The viruses were enumerated using an agar overlay method. TYE agar (TYE plus 10 g/L agar) plates and fresh cultures of host bacteria were prepared in advance of testing each day.

Just prior to use, 1 mL of a sterile stock solution of calcium chloride (22 g/L) was added to each 20 mL of host bacteria culture. Three drops of the bacteria culture, to provide a lawn of host cells, and 0.1 mL of a proper dilution of virus suspension were then added to 3 mL of soft TYE agar. This mixture was poured over a prepared hard agar plate. Virus plaques were counted after an 8 to 24 hour incubation at 35°C .

MIXED SECOND-ORDER KINETICS

The kinetics of inactivation are assumed to be first order with respect to the surviving organism density and first order with respect to light intensity. For any point, p, in the reactor,

$$R = KI_p N \tag{1}$$

where

- R = disinfection rate (organisms/cm³ · sec),
- K = rate constant ($\mu\text{w sec/cm}^2$)⁻¹,
- I_p = intensity at p ($\mu\text{w/cm}^2$), and
- N = surviving organism density (organisms/cm³).

For the batch reactor used with the conditions discussed, that is, a flat tray reactor with a thick film of high UV transmission, then the intensity is uniform and

$$N/N_0 = \exp(-KIpt) \tag{2}$$

where

- N = density of surviving organisms (cfu or pfu/cm³),
- N₀ = initial density, and
- t = length of exposure (sec).

For a completely mixed flow-through reactor of annular design, the intensity at all points is not uniform. The light intensity at p may be estimated from Lambert's law for light that is directed radially from an infinitely long line source,¹⁵

$$\frac{1}{r} \frac{d(rI_p)}{dr} = -EI_p \tag{3}$$

where

- E(cm⁻¹) = extinction coefficient at 254 nm expressed in base e, and
- r̄ = radial distance (cm) to p.

Comparison of this approach for estimating the light intensity profile¹⁶ to the finite length lamp model¹⁷ concluded that for most annular reactor configurations the average intensity calculated using the radial model was not different enough from that from the finite length lamp model to warrant the use of the more complex light profile model. The solution of Equation 3 for the boundary conditions $I_p = I_0$ at $r = r_0$, is

$$I_p = \left(\frac{r_0 I_0}{r} \right) \exp(-E(r - r_0)) \quad (4)$$

where

r_0 = radial distance from the center of the lamp to the outside of the quartz tube (cm), and

I_0 = light intensity at the quartz tube surface ($\mu\text{W}/\text{cm}^2$)

Equation 5 is a steady state surviving-organism material balance over the reactor volume,

$$N_0 Q - N Q = \int R dV \quad (5)$$

where

V = volume of the reactor (cm^3), and

Q = flow rate (cm^3/sec).

Substitution of Equations 1 and 4 and specification of the volume integral transforms Equation 5 into the following equation:

$$(N_0 - N)Q = \int_0^L \int_0^{2\pi} \int_{r_0}^{\bar{r}} \left(\frac{K N I_0 r_0}{r} \right) \exp(-E[r - r_0]) r dr d\theta dl \quad (6)$$

where

θ = angle about the line source,

l = the variable for the length of the reactor (cm),

L = length of the reactor (cm), and

$\bar{r}$ = radius of the reactor.

The solution of Equation 6 is

$$(N_0 - N)Q = \frac{2\pi K N I_0 r_0 L}{E} (1 - \exp[-E(\bar{r} - r_0)]) \quad (7)$$

which can be simplified to

$$\frac{N}{N_0} = \frac{1}{1 + m K I_0 \tau} \quad (8)$$

where

τ = theoretical retention time in the reactor (sec), V/Q

V , reactor volume (cm^3) = $\pi(\bar{r}^2 - r_0^2)L$, and (9)

m is a dimensionless function of reactor geometry and water quality and represents the ratio of the average light intensity in the reactor to the intensity at the surface of the quartz tube,

$$\frac{2 r_0 (1 - \exp[-E(\bar{r} - r_0)])}{E(\bar{r}^2 - r_0^2)} \quad (10)$$

The relation between m and base 10 absorbance, A , is shown in Figure 2, where,

$$E = 2.303A. \quad (11)$$

The form of Equation 8 is typical of a solution for a first order reaction in a completely mixed, flow-through reactor. The result is the same as that derived by Haas and Sakellaropoulos,¹¹ who defined an average reaction rate based on the average light intensity. The derivation here is based on a point reaction rate averaged over the reactor volume.

RESULTS

Inactivation results from the batch reactor are presented in Figure 3. Only the f2 virus shows a linear relation between the logarithm of the survival fraction and contact time, which is indicative of mixed second-order kinetics. Inactivation of 99.9% of f2 is observed after 165 seconds. The plots for *E. coli* and *C. parapsilosis* are not straight lines, indicating a poor relation to mixed second-order kinetics. The curve plotted from *E. coli* data exhibits a resistant "shoulder" extending over 7 to 10 seconds contact followed by a sharp increase in disinfection efficiency, such that 99.9% inactivation is achieved after about 35 seconds. A resistant "shoulder" of 40 to 60 seconds of contact is observed in the data for *C. parapsilosis*. In fact, *C. parapsilosis* is more resistant than f2 virus in the first 45 seconds of contact; however, 99.9% of *C. parapsilosis* is inactivated after about 80 seconds as compared to 165 seconds required to inactivate 99.9% of the f2 virus.

A common method of comparing the inactivation response of different organisms is to report the values of intensity times, contact time, or dose ($\mu\text{W}/\text{sec} \cdot \text{cm}^2$) of UV in batch tests required to inactivate a given fraction of the initial population. The required dose to inactivate 99% and 99.9% of the three organisms in the batch reactor are presented in Table 2. Comparison of the values for the different organisms indicates that the order of resistance is f2 virus > *C. parapsilosis* > *E. coli*.

Assuming that mixed second-order kinetics are adequate to describe inactivation, best-fit linear regression kinetic constants from the linear sections of the batch inactivation plots are $K = 0.893 \times 10^{-3} \text{ cm}^2/\mu\text{W} \cdot \text{sec}$ for *E. coli*, 0.0724×10^{-3} for f2 virus, and 0.433×10^{-3} for *C. parapsilosis*. This type of analysis also indicates that the relative resistance is f2 virus > *C. parapsilosis* > *E. coli* in batch reactions.

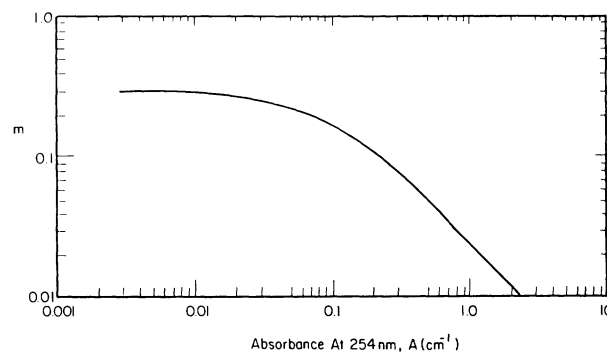


Figure 2—Absorbance factor, m , in the UV reactor.

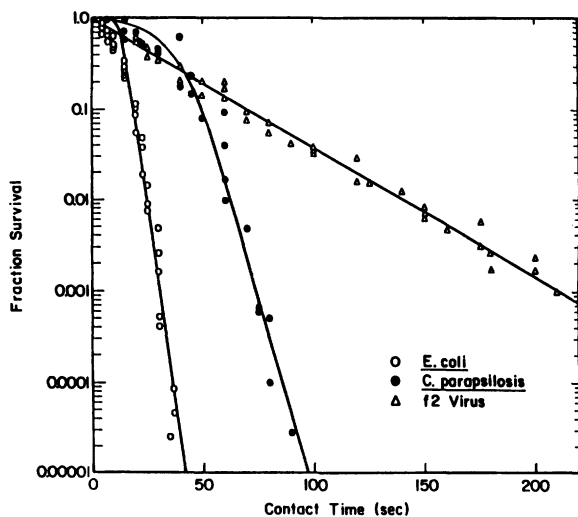


Figure 3—Batch inactivation data.

Inactivation data from the completely mixed flow-through reactor are presented in Figures 4, 5, and 6. These data are representative of the range of absorbances tested.¹⁶ The solid lines are predictions from mixed second-order kinetic constants obtained from results in the flow-through reactor using water with no PHBA added (calibrated constants; data not presented here). From a first glance, it seems as if mixed second-order kinetics hold well for the completely mixed flow-through reactor. Mixed second-order kinetic constants calculated from available data for the flow-through reactor are $0.1325 \times 10^{-3} \text{ cm}^2/\mu\text{w} \cdot \text{sec}$ for *E. coli*, 0.09229×10^{-3} for f2 virus and 0.04553×10^{-3} for *C. parapsilosis*. Based on this analysis of results from the flow-through reactor, the relative resistance of the organisms is *C. parapsilosis* > f2 virus > *E. coli*. This represents an inversion in the apparent resistances in the batch data where f2 virus > *C. parapsilosis* > f2 virus.

DISCUSSION

Even though linear portions on batch curves are evident for all three organisms, the kinetic constant from the linear section cannot be used to predict the behavior of the completely mixed flow-through reactor if a resistant “shoulder” region exists. First, the relative resistance of the organisms changed in the two reactors. In batch experiments the resistance order was f2 virus > *C. parapsilosis* > *E. coli* and in the completely mixed flow-through reactor the order was *C. parapsilosis* > f2 virus > *E. coli*. This is explained by the initial resistance of *C. parapsilosis*

Table 2—Dose response of test organisms in batch reactions.

	Dose required ($\mu\text{w}/\text{sec} \cdot \text{cm}^2$) for inactivation	
	99%	99.9%
<i>E. coli</i>	1.22×10^3	1.40×10^3
f2 virus	5.00×10^3	7.29×10^3
<i>C. parapsilosis</i>	3.29×10^3	3.60×10^3

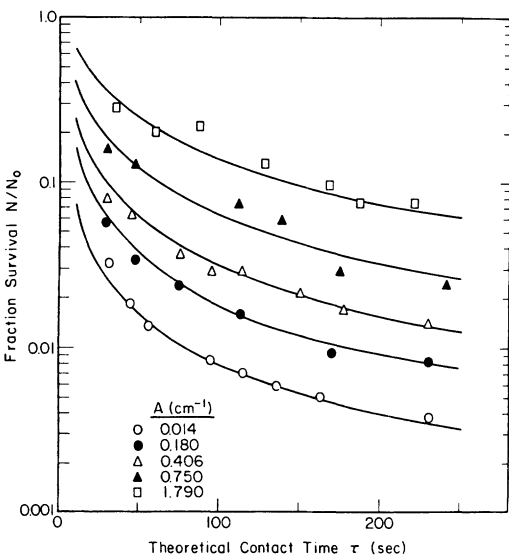


Figure 4—Inactivation of *E. coli* in the completely mixed, flow-through reactor.

and the expression of this resistance in a completely mixed flow-through reactor. Second, the prediction of the inactivation of f2 virus is quite good between reactor systems; *E. coli* and *C. parapsilosis* inactivation in the completely mixed flow-through reactor is inconsistent with predictions from batch data. This is shown in Figure 7 where the fraction survival in the completely mixed reactor is plotted against the dimensionless group $\text{mkl}_0\tau$. The solid line is the prediction from Equation 8; the plotted data are normalized with the mixed second-order kinetic constants from batch data. The prediction of f2 inactivation is quite good with a coefficient of determination $R^2 = 0.97$. The data of *C. parapsilosis* and *E. coli*, however, lie far to the right of the predicted line. Thus, these two organisms seem more resistant in the flow-through reactor than would be predicted using batch kinetic constants.

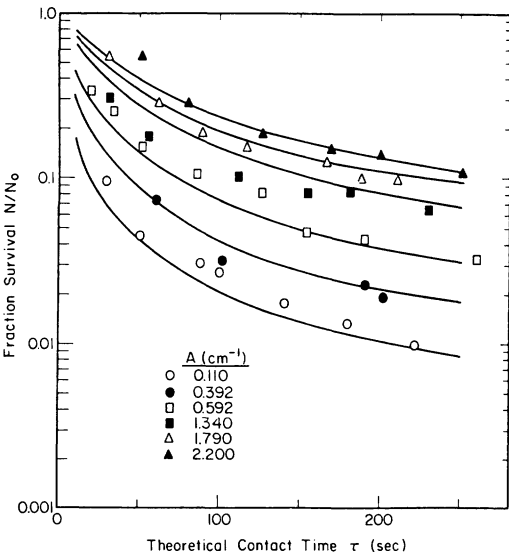


Figure 5—Inactivation of f2 virus in the completely mixed, flow-through reactor.

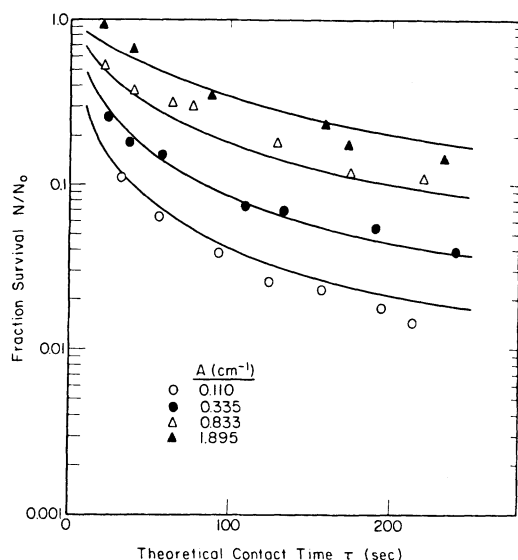


Figure 6—Inactivation of *C. parapsilosis* in the completely mixed, flow-through reactor.

This analysis should serve as a warning to researchers who analyze kinetic behavior in completely mixed flow-through reactors without verification from batch reactions. From the trend in the normalized plot (Figure 7), it is clear that the *E. coli* and the *C. parapsilosis* data could be mistaken for verification of a mixed second-order model if incorrect (low) kinetic constants were used, whereby the data would fit closely to the predicted line. This error cannot be observed unless the kinetic behavior is cross-matched with batch data.

It is common practice¹⁸ to report the resistances of organisms to disinfection by a "concentration-time" or "dose-time" number that results in a predetermined inactivation level. This can be interpreted as a pseudo-mixed, second-order approach. This discussion suggests that (1) this method of analysis cannot be used to extrapolate to reactor systems with complex mixing regimes, (2) this data analysis method will result in different dose numbers depending on which type of reactor system is used by various researchers, and (3) this method can give erroneous rankings of organism resistance if the only data available are from completely mixed flow-through reactors.

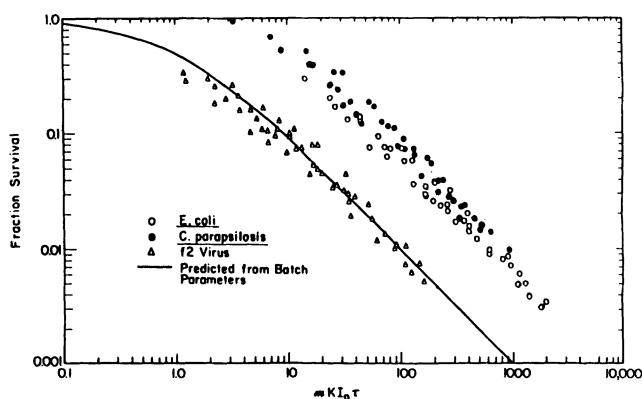


Figure 7—Prediction of inactivation in the flow-through reactor using batch kinetic constants.

CONCLUSIONS

Mixed second-order kinetic models are adequate for organisms that do not exhibit initial resistance to a disinfectant, as exemplified by f2 virus in the presence of UV light. A kinetic constant from batch data was useful to predict inactivation in a completely mixed flow-through reactor. Mixed second-order kinetic models are not adequate for organisms that exhibit initial resistance, such as *E. coli* and *C. parapsilosis*. Kinetic constants from linear sections of the batch inactivation curves were not useful to predict inactivation in a completely mixed flow-through reactor.

The failure of mixed second-order kinetics seems to result from an inability to translate initial resistance phenomenon between different reactor configurations. Other kinetic expressions¹⁹ that include the treatment of the initial resistance are more suitable for the analysis of inactivation data, especially when the desired outcome is the design of reactor systems with complex mixing. The use of a completely mixed flow-through reactor for the study of inactivation kinetics is discouraged unless supportive data from batch reactors also is available. The completely mixed flow-through reactor seems to obscure the initial resistance phenomenon. The error can be so great that it alters the apparent order of resistance of various organisms.

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Authors. Blaine F. Severin is an environmental engineer, Tennessee Eastman Co., Kingsport, Tenn. Makram T. Suidan is an associate professor, Bruce E. Rittmann is an assistant professor, and Richard S. Engelbrecht is a professor, Department of Civil Engineering, University of Illinois at Urbana-Champaign. Correspondence should be sent to: Blaine F. Severin, Tennessee Eastman Co., Box 511, Kingsport, TN 37662.

REFERENCES

1. Johnson, J. D., *et al.*, "UV Disinfection of Secondary Effluent," Proceedings, National Symposium on Progress in Wastewater Disinfection. EPA-600/7-79-018, U. S. EPA, Cincinnati, Ohio, September (1978).
2. Jolley, R. L., *et al.*, "Effects of Chlorine, Ozone, and Ultraviolet light on Nonvolatile Organics in Wastewater Effluents." Proceedings, National Symposium on Progress in Wastewater Technology. EPA-600/7-79-018, U. S. EPA Cincinnati, Ohio, September (1978).
3. Luckiesh, M., and Holladay, L. L., "Disinfecting Water by Means of Germicidal Lamps," *General Electric Review*, **47**, 45 (1944).
4. Oliver, B. G., and Carey, J. H., "Ultraviolet disinfection an alternative to chlorination," *J. Water Pollut. Control Fed.*, **48**, 2619 (1976).
5. Oliver, B. G., and Cosgrove, E. G., "The Disinfection of Sewage Treatment Plant Effluents Using Ultraviolet Light." *Can. J. Chem. Eng.*, **53**, 170 (1975).
6. Roeber, J. A., and Hoot, F. M., "Ultraviolet Disinfection of Activated Sludge Effluent Discharging to Shellfish Waters." Environmental Protection Technology Series, EPA-WPRD 139-01-68, U. S. EPA, Washington, D. C. December (1975).
7. Scheible, O. K., Binkowski, G., and Milligan, T. J., "Full Scale Evaluation of UV Disinfection of a Secondary Effluent." Proceedings, National Symposium on Progress in Wastewater Disinfection. EPA-600/7-79-018, U. S. EPA, Cincinnati, Ohio, September (1978).

8. Severin, B. F., "Disinfection of municipal wastewater effluents with UV light." *J. Water Pollut. Control Fed.*, **52**, 7, 2007 (1980).
9. Yip R. W., and Konasewich, D. E., "Ultraviolet Sterilization of Water—Its Potential and Limitations." *Water Pollut. Control (Can.)*, **14**, June, (1972).
10. Cortelyou, J. R., *et al.*, "Effects of Ultraviolet Irradiation on Large Populations of Certain Water-Borne Bacteria in Motion." *Appl. Microbiol.* **2**, 227 (1954).
11. Haas, C. N., and Sakellaropoulos, G. P., "Rational Analysis of UV Disinfection Reactors." ASCE National Conference on Environmental Engineers, San Francisco, Cal., July, (1979).
12. Calvert, J. G., and Pitts, J. N., "Photochemistry." John Wiley & Sons, Inc., New York (1966).
13. Engelbrecht, R. S., *et al.*, "Acid-Fast Bacteria and Yeasts as Indicators of Disinfection Efficiency." EPA-600/2-79-091, U. S. EPA, Washington, D. C. (1979).
14. Leob, T., and Zinder, N. "A Bacteriophage Containing RVA." *Proc. Natl. Acad. Sci., U. S.*, **47**, 282 (1961).
15. Jacob, S. M., and Dranoff, J. S., "Radial Scale-up of Perfectly Mixed Photochemical Reactors." *Chem. Eng. Progress Symposium Series*, **62**, 68 (1966).
16. Severin, B. F., "Kinetic Modeling of Microbial Inactivation by Ultraviolet Light." Ph.D. Thesis, Dept. Civil Eng., University of Illinois, Urbana, Ill. (1982).
17. Jacob, S. M., and Dranoff, J. S. "Light Intensity Profiles in a Perfectly Mixed Photoreactor." *AIChE J.*, **16**, 3, 359–363 (1970).
18. National Academy of Sciences, Drinking Water and Health, Vol. 2, Safe Drinking Water Committee, National Academy Press, Washington, D. C. (1980).
19. Severin, B. F., Suidan, M. T., and Engelbrechet, R. S., "Kinetic Modeling of UV Disinfection of Water." *Water Res.*, **17**, 1669 (1983).